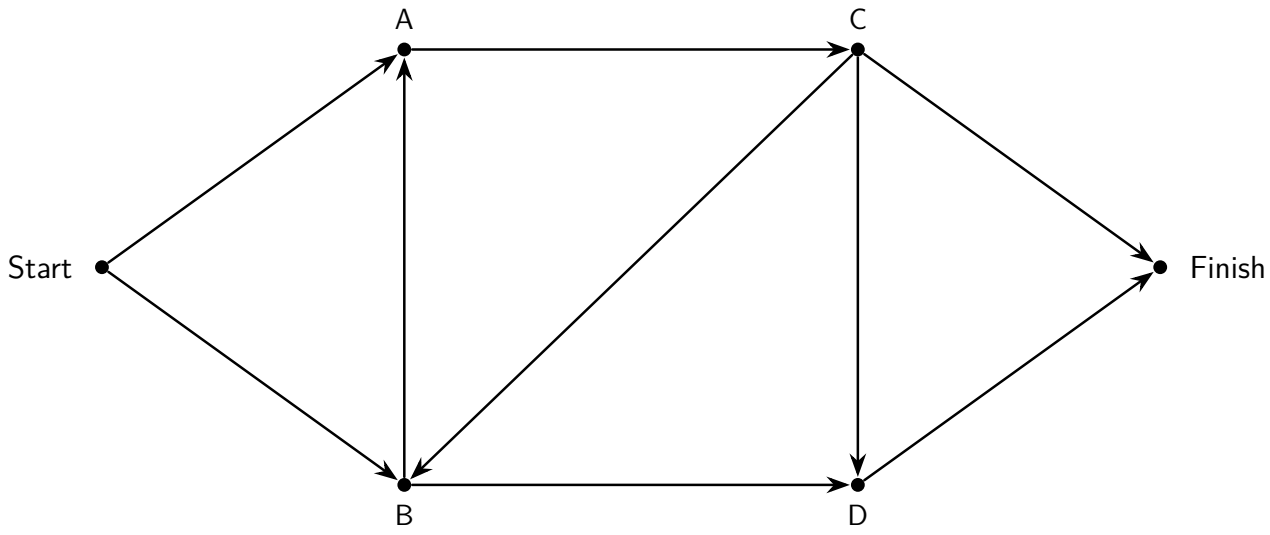
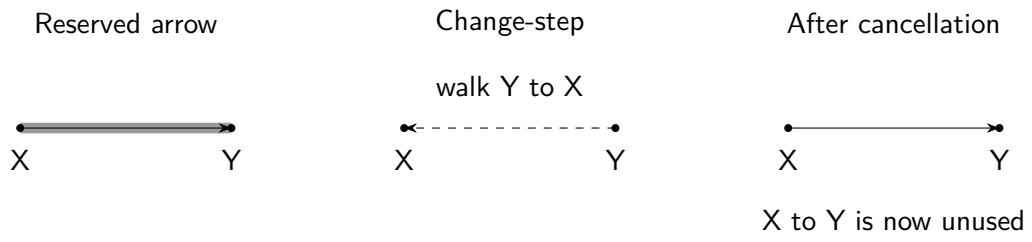
A route follows arrows from Start to Finish without visiting a dot twice. Routes may meet at dots, but each arrow can belong to only one route in a collection. Use a different color or line pattern for each route. A marker closes one arrow.

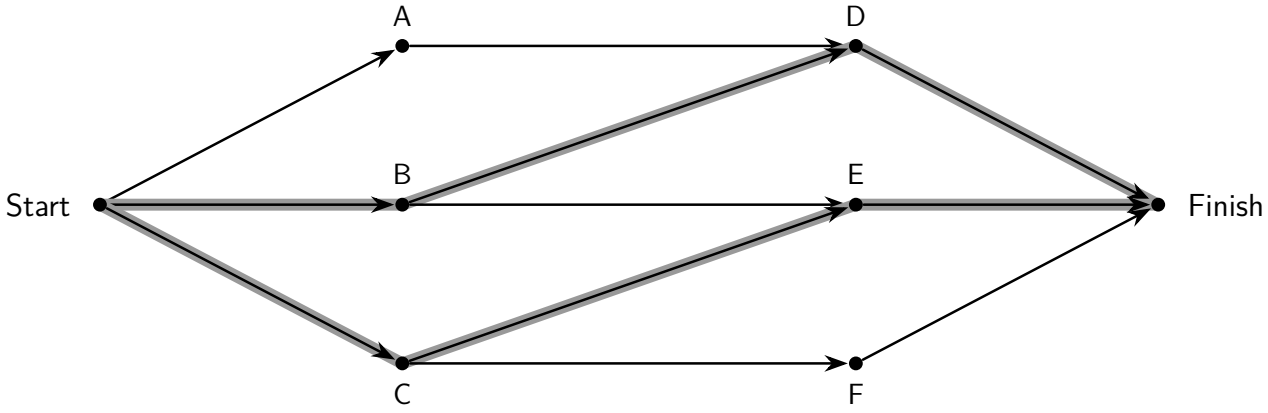
Problem 4: Find a smallest set of closing arrows and a route that uses two of those arrows. Can that route belong to a largest collection on this board? Explain.

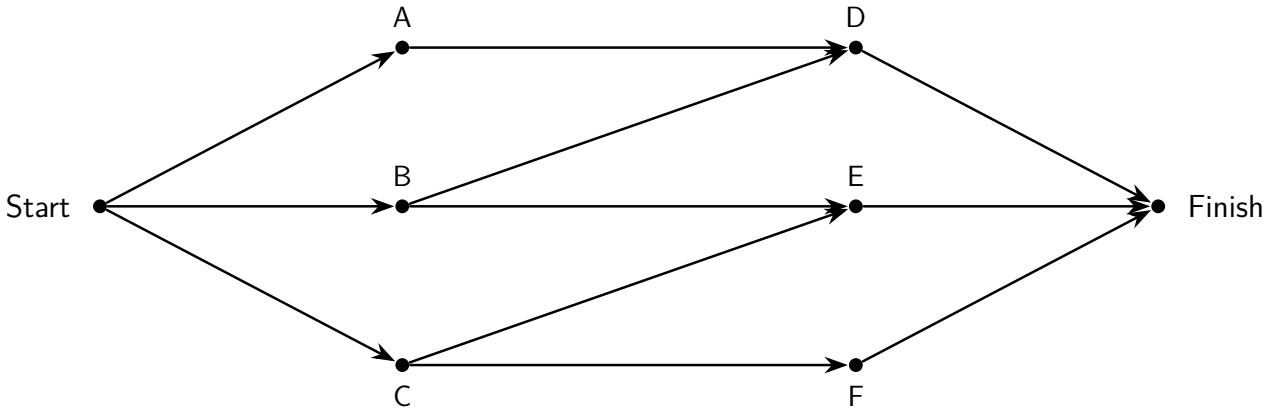


A change-walk follows an unused arrow forward or a reserved arrow backward. A forward step reserves its arrow; a backward step cancels its reservation. A change-walk visits each dot at most once. Final routes follow the printed arrows.

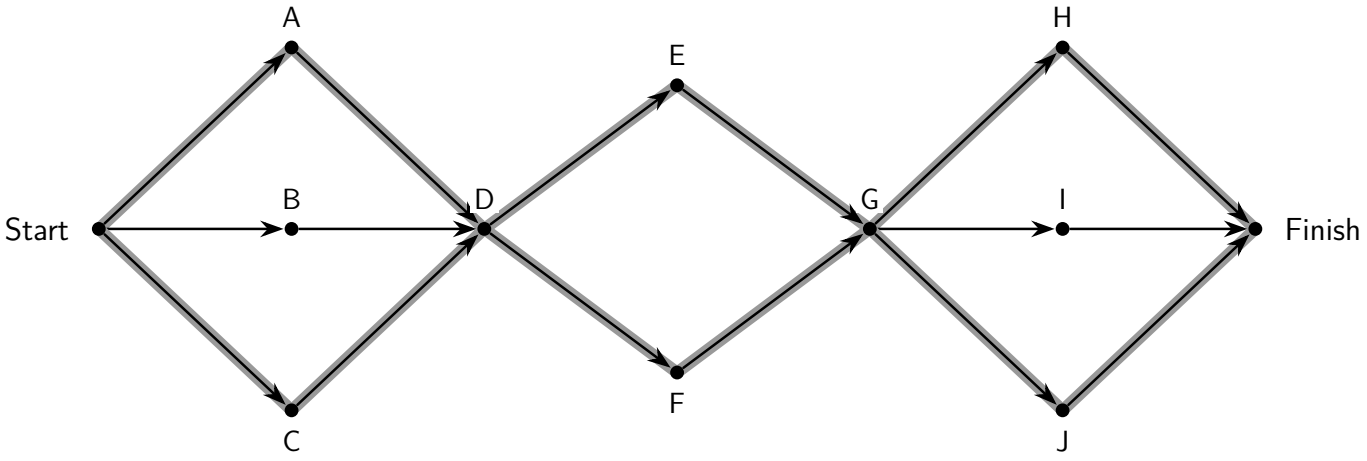


Problem 5: Find a change-walk from Start to Finish that increases the number of routes by one, on this board and on both boards of Problem 1. Draw each resulting route collection.





Problem 6: The thick arrows form two reserved routes. Find every dot you can reach from Start by change-walk steps. Find a smallest set of closing arrows, and explain why it matches the route collection.



Problem 7: On any finite board with these rules, suppose no change-walk from Start can reach Finish. Must the reserved route collection already be largest? Explain why, or draw a board where it is not.